STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-COMPOSE-TENSOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5005-5030 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 5032-5082, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_sdga.json (pb=1837, stmt 5005-5030, proof 5032-5082).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FTK: sources/stacks-sdga/pinned-original-source.tex, source lines 4977–4986. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSH: sources/stacks-sdga/pinned-original-source.tex, source lines 3186–3196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FTJ: sources/stacks-sdga/pinned-original-source.tex, source lines 4946–4951. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$, $\mathcal{A}'$, $\mathcal{A}''$ be differential graded $\mathcal{O}$-algebras. Let $\mathcal{N}$ and $\mathcal{N}'$ be a differential graded $(\mathcal{A}, \mathcal{A}')$-bimodule and $(\mathcal{A}', \mathcal{A}'')$-bimodule. Assume that the canonical map*

$$\mathcal{N} \otimes^{\mathbf{L}}_{\mathcal{A}'} \mathcal{N}' \longrightarrow \mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}'$$

in $D(\mathcal{A}'', d)$ is a quasi-isomorphism. Then we have

$$(\mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{N}) \otimes^{\mathbf{L}}_{\mathcal{A}'} \mathcal{N}' = \mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} (\mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}')$$

as functors $D(\mathcal{A}, d) \to D(\mathcal{A}'', d)$.

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Proof. Choose a good differential graded $\mathcal{A}$-module $\mathcal{P}$ and a quasi-isomorphism $\mathcal{P} \to \mathcal{M}$, see Lemma 0FSH. Then

$$\mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} (\mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}') = \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}'$$

and we have

$$(\mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{N}) \otimes^{\mathbf{L}}_{\mathcal{A}'} \mathcal{N}' = (\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}) \otimes^{\mathbf{L}}_{\mathcal{A}'} \mathcal{N}'$$

Thus we have to show the canonical map

$$(\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}) \otimes^{\mathbf{L}}_{\mathcal{A}'} \mathcal{N}' \longrightarrow \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}'$$

is a quasi-isomorphism. Choose a quasi-isomorphism $\mathcal{Q} \to \mathcal{N}'$ where $\mathcal{Q}$ is a good left differential graded $\mathcal{A}'$-module (Lemma 0FSH). By Lemma 0FTK the map above as a map in the derived category of $\mathcal{O}$-modules is the map

$$\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \otimes_{\mathcal{A}'} \mathcal{Q} \longrightarrow \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}'$$

Since $\mathcal{N} \otimes_{\mathcal{A}'} \mathcal{Q} \to \mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}'$ is a quasi-isomorphism by assumption and $\mathcal{P}$ is a good differential graded $\mathcal{A}$-module this map is an quasi-isomorphism by Lemma 0FTJ (the left and right hand side compute $\mathcal{P} \otimes^{\mathbf{L}}_{\mathcal{A}} (\mathcal{N} \otimes_{\mathcal{A}'} \mathcal{Q})$ and $\mathcal{P} \otimes^{\mathbf{L}}_{\mathcal{A}} (\mathcal{N} \otimes_{\mathcal{A}'} \mathcal{N}')$ or you can just repeat the argument in the proof of the lemma). □